ITI RESOURCE HUB - LESSON PLAN

Trade: ICTSM | Topic: Motherboard Form Factors & POST Diagnostics

1. Instructional Objectives: Trainee will be able to identify ATX, Micro-ATX, and Mini-ITX form factors.
2. Instructional Aids: Motherboard display unit, POST diagnostic PCI card, Multimeter, Screwdriver set.
3. Motivation / Review: Explain common symptoms when a computer powers on without video display.
4. Presentation Points: Chipset architecture (Northbridge / Southbridge or PCH), bus lanes, BIOS beep codes.
5. Practical Application: Trainees insert diagnostic card into PCIe slot and record hexadecimal error codes.
6. Safety Precaution: Ground wrist with anti-static ESD strap before handling delicate microchips.
7. Evaluation & Assignment: Complete the fault diagnosis questionnaire and test 3 faulty test boards.